Homework: The Biological Learning Rules

Hebbian Learning · Predictive & Efficient Coding · Reinforcement Learning

Companion to Lectures 3–5 (Biological Learning Rules I & II)

Due: June 22nd Total: 150 pts

These three problem sets accompany the lectures on biological learning rules. Each is designed for roughly one to two weeks of work and combines analytical derivations with conceptual questions linking the mathematics to neurobiology; the numerical suggestions at the end can be used as optional extensions. Notation follows the lectures: a linear neuron computes $y = \mathbf{w}^\top \mathbf{x}$, the input correlation (second-moment) matrix is $\mathbf{C} = \langle \mathbf{x}\mathbf{x}^\top \rangle$, and η, α denote learning rates.

Part 1 — Hebbian Learning & Synaptic Plasticity

Topics: Oja's rule and online PCA, the neuron as a dimensionality-reduction device, spike-timing-dependent plasticity.

Problem 1. Oja's rule learns the first principal component [17 pts]

The plain Hebbian rule $\Delta\mathbf{w} = \eta\, y\, \mathbf{x}$ is local but unstable: with $y = \mathbf{w}^\top \mathbf{x}$ its averaged dynamics are $\tau\dot{\mathbf{w}} = \mathbf{C}\mathbf{w}$, which drive $\|\mathbf{w}\| \to \infty$ along the leading eigenvector of $\mathbf{C} = \langle \mathbf{x}\mathbf{x}^\top \rangle$, so it requires normalization. Oja's rule supplies exactly such a self-normalizing term, and the lecture flags the following as a homework derivation:

$$\tau_w\, \dot{\mathbf{w}} \;=\; y\, \mathbf{x} \;-\; y^2\, \mathbf{w}, \qquad y = \mathbf{w}^\top \mathbf{x}.$$

(a) Show that the averaged dynamics are $\tau_w\, \dot{\mathbf{w}} = \mathbf{C}\mathbf{w} - (\mathbf{w}^\top \mathbf{C}\mathbf{w})\, \mathbf{w}$.

(b) Compute $\dfrac{d\|\mathbf{w}\|^2}{dt}$ and show it equals $\frac{2}{\tau_w}\, y^2\big(1 - \|\mathbf{w}\|^2\big)$ for the instantaneous rule. For the averaged dynamics this becomes $\frac{2}{\tau_w}(\mathbf{w}^\top \mathbf{C}\mathbf{w})\big(1 - \|\mathbf{w}\|^2\big)$, equivalently $\propto \|\mathbf{w}\|^2(1 - \|\mathbf{w}\|^2)$ for a fixed weight direction. Conclude that the norm is driven toward $\|\mathbf{w}\| = 1$, so the weight vector is *bounded* — curing the runaway growth of the plain Hebbian rule $\Delta\mathbf{w} = \eta\, y\, \mathbf{x}$.

(c) Assuming $\mathbf{C}$ is positive definite, show that the nonzero fixed points of the averaged dynamics are the eigenvectors of $\mathbf{C}$ with $\|\mathbf{w}\| = 1$ (there is also the trivial fixed point $\mathbf{w} = \mathbf{0}$). If the largest eigenvalue is unique, use linear stability analysis to show the *only stable* nonzero fixed points are $\pm\mathbf{e}_1$. Hence a single neuron with this local, Hebbian-like rule performs online PCA. (If $\mathbf{C}$ has zero eigenvalues or degenerate leading eigenvalues, state how the fixed-point set changes.)

(d) Connect to coding: for centered inputs, show that projecting onto $\mathbf{e}_1$ minimizes the linear reconstruction error $\sum_t \|\mathbf{x}_t - \mathbf{w}y_t\|^2$ over unit $\mathbf{w}$. In one or two sentences, state why "the neuron as a dimensionality-reduction device" is the unifying idea here.

Problem 2. Spike-timing-dependent plasticity [15 pts]

Pair-based STDP modifies a synapse as a function of $\Delta t = t_{\text{post}} - t_{\text{pre}}$:

$$\Delta w(\Delta t) = \begin{cases} A_+ \, e^{-\Delta t/\tau_+}, & \Delta t > 0 \quad \text{(pre before post: LTP)} \\ -A_- \, e^{\Delta t/\tau_-}, & \Delta t < 0 \quad \text{(post before pre: LTD).} \end{cases}$$

(a) Sketch the window and explain in one sentence how its temporal asymmetry encodes a notion of *causality* (and thus temporal credit assignment).

(b) Suppose pre- and postsynaptic spikes are independent Poisson processes at rates $r_{\text{pre}}, r_{\text{post}}$, so that the density of pairs with lag Δt is $r_{\text{pre}} r_{\text{post}}$ per unit time per unit lag. Show that the mean weight change per unit time is

$$\left\langle \frac{dw}{dt} \right\rangle = r_{\text{pre}} \, r_{\text{post}} \left(A_+ \tau_+ - A_- \tau_- \right),$$

and state the condition on $(A_+\tau_+,\, A_-\tau_-)$ for the synapse to be, on average, potentiating, depressing, or balanced.

(c) Show that for independent Poisson spikes the leading rate-dependent part of STDP is a Hebbian rate product $\propto r_{\text{pre}} r_{\text{post}}$. What additional ingredient — for example, nontrivial pre–post temporal correlations or stimulus-locked spike timing — must be added for STDP to do something a pure rate rule cannot?

Part 2 — Predictive Coding

Topics: Reconstruction and PCA, sparse coding as MAP inference, the inference/learning circuit and its (non)locality, two-level predictive coding (Rao–Ballard), predictive coding as an approximation to backpropagation, and feedback alignment for the weight-transport problem.

Problem 1. From reconstruction to a population code [13 pts]

A linear population code encodes $\mathbf{y} = \mathbf{W}^\top \mathbf{x}$ and decodes $\hat{\mathbf{x}} = \mathbf{W}\mathbf{y}$, with $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{y} \in \mathbb{R}^k$.

(a) For $k < n$ and the orthonormality constraint $\mathbf{W}^\top \mathbf{W} = \mathbf{I}_k$, show that minimizing $\sum_t \|\mathbf{x}_t - \mathbf{W}\mathbf{W}^\top \mathbf{x}_t\|^2$ is solved by taking the columns of $\mathbf{W}$ to span the top-k eigenspace (principal subspace) of $\mathbf{C} = \langle \mathbf{x}\mathbf{x}^\top \rangle$. (You may build on Problem 1 of PS1.)

(b) Contrast the PCA solution (orthogonal, distributed, $k \leq n$) with a sparse, *overcomplete* code ($k > n$). Which objective term changes, and give two reasons — one metabolic, one representational — the cortex might favor sparse overcomplete codes (e.g. V1 simple cells).

Problem 2. Sparse coding as MAP inference (Olshausen & Field) [18 pts]

Take the generative model $\mathbf{x} = \mathbf{W}\mathbf{y} + \boldsymbol{\eta}$ with $\boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \sigma_\eta^2 \mathbf{I})$ and the factorized Laplace prior $p(\mathbf{y}) \propto e^{-\lambda \|\mathbf{y}\|_1}$.

(a) Write $\log p(\mathbf{y} \mid \mathbf{x})$ up to constants and show that MAP inference is the LASSO problem $\min_{\mathbf{y}} \frac{1}{2\sigma_\eta^2}\|\mathbf{x} - \mathbf{W}\mathbf{y}\|_2^2 + \lambda \|\mathbf{y}\|_1$.

(b) Performing gradient descent on the energy, $\dot{\mathbf{y}} = -\partial E / \partial \mathbf{y}$, derive $\dot{\mathbf{y}} \propto \mathbf{W}^\top(\mathbf{x} - \mathbf{W}\mathbf{y}) - \lambda \operatorname{sign}(\mathbf{y})$ and identify the three terms as: feedforward drive, recurrent "explaining-away," and a sparsity threshold.

(c) The lecture writes the circuit as $\dot{\mathbf{y}} = -\lambda_2 \mathbf{y} + \mathbf{W}^\top \mathbf{x} - \mathbf{W}^\top \mathbf{W}\, \mathbf{y}$. Which prior on $\mathbf{y}$ does this dynamics implement (Gaussian or Laplace), and what is the function of the $-\mathbf{W}^\top \mathbf{W}\, \mathbf{y}$ term anatomically (hint: recurrent / lateral inhibition; competition between features)?

(d) Derive the synaptic learning rule by gradient descent on the same E with respect to $\mathbf{W}$ (at the inference fixed point), obtaining $\dot{W}_{ij} \propto x_i y_j - y_j \sum_k W_{ik} y_k$. If one adds an explicit weight-decay penalty $\frac{\beta}{2}\|\mathbf{W}\|_F^2$, show that this becomes $\dot{W}_{ij} \propto -\beta W_{ij} + x_i y_j - y_j \sum_k W_{ik} y_k$. Explain why the reconstruction term makes the rule *non-local*, contrast with Oja's strictly local rule, and propose one biophysical mechanism that could supply the missing information.

Problem 3. Two-level predictive coding (Rao & Ballard) [17 pts]

Consider one cortical level with representation $\mathbf{r}$, a top-down prediction $\mathbf{W}\mathbf{r}$ of the input $\mathbf{x}$, and a top-down prior expectation $\mathbf{r}_{\text{td}}$ for $\mathbf{r}$. Define the energy

$$E = \frac{1}{2\sigma_x^2}\|\mathbf{x} - \mathbf{W}\mathbf{r}\|^2 + \frac{1}{2\sigma_r^2}\|\mathbf{r} - \mathbf{r}_{\text{td}}\|^2.$$

(a) Let the prediction-error signal be $\mathbf{e} = \mathbf{x} - \mathbf{W}\mathbf{r}$. Derive the inference dynamics $\dot{\mathbf{r}} = -\partial E/\partial \mathbf{r}$ and show it has the form $\dot{\mathbf{r}} \propto \mathbf{W}^\top \mathbf{e} - \frac{\sigma_x^2}{\sigma_r^2}(\mathbf{r} - \mathbf{r}_{\text{td}})$: bottom-up error drives $\mathbf{r}$ up, the prior pulls it toward the top-down prediction.

(b) Argue that the same weight matrix appears as $\mathbf{W}$ (top-down prediction) and $\mathbf{W}^\top$ (bottom-up error routing). Explain how this bears on the *weight-transport problem* that makes backpropagation biologically suspect — does predictive coding solve it, assume it away, or replace it with a symmetry assumption? Discuss briefly.

(c) Derive the learning rule for $\mathbf{W}$ as gradient descent on E at the inference fixed point, and show it is *Hebbian and local*: $\Delta\mathbf{W} \propto \mathbf{e}\,\mathbf{r}^\top$, i.e. the product of a prediction-error unit and a representation unit. Why is this the central biological-plausibility payoff of predictive coding?

(d) Conceptual. State (no proof needed) the sense in which hierarchical predictive coding can approximate backpropagation, and propose a mapping of "prediction-error" vs. "representation" units onto cortical laminar microcircuitry (e.g. superficial vs. deep pyramidal populations). What experimental signature would distinguish error units from representation units?

Problem 4. Predictive coding, backpropagation, and weight transport [19 pts]

Consider an L-layer feedforward network with pre-activations $\mathbf{z}^{(l)} = \mathbf{W}^{(l)}\mathbf{a}^{(l-1)}$ and activations $\mathbf{a}^{(l)} = f(\mathbf{z}^{(l)})$, input $\mathbf{a}^{(0)} = \mathbf{x}$, output $\mathbf{a}^{(L)}$, and a supervised loss $\mathcal{L}(\mathbf{a}^{(L)}, \mathbf{t})$ against target $\mathbf{t}$. ($\odot$ denotes elementwise product.)

(a) *Backprop and its sore spot.* Write the backpropagation recursion for the error $\boldsymbol{\delta}^{(l)} \equiv \partial\mathcal{L}/\partial\mathbf{z}^{(l)}$, namely $\boldsymbol{\delta}^{(L)} = f'(\mathbf{z}^{(L)}) \odot \partial\mathcal{L}/\partial\mathbf{a}^{(L)}$ and $\boldsymbol{\delta}^{(l)} = f'(\mathbf{z}^{(l)}) \odot \left(\mathbf{W}^{(l+1)\top}\boldsymbol{\delta}^{(l+1)}\right)$, together with the gradient $\partial\mathcal{L}/\partial\mathbf{W}^{(l)} = \boldsymbol{\delta}^{(l)}\mathbf{a}^{(l-1)\top}$. Identify the one factor that is biologically problematic and name the problem in a phrase.

(b) *Predictive-coding counterpart.* Build the predictive-coding network in a preactivation-like convention: each layer carries representation units $r^{(l)}$ with activity $a^{(l)} = f(r^{(l)})$, and prediction-error units

$$\varepsilon^{(l)} = r^{(l)} - \mathbf{W}^{(l)} f\big(r^{(l-1)}\big),$$

with the input clamped ($f(r^{(0)}) = \mathbf{x}$) and the output gently nudged toward $\mathbf{t}$. Inference relaxes the $r^{(l)}$ by gradient descent on $E = \frac{1}{2}\sum_l \|\varepsilon^{(l)}\|^2$ (the two-phase scheme of the lecture). Write $\dot{r}^{(l)} = -\partial E/\partial r^{(l)}$ and show it balances the local error $-\varepsilon^{(l)}$ against a feedback term $f'(r^{(l)}) \odot \left(\mathbf{W}^{(l+1)\top}\varepsilon^{(l+1)}\right)$. (You may quote the single-level result of Problem 3 in this part rather than re-deriving it.)

(c) *Equivalence (Whittington & Bogacz, 2017).* Argue that at the inference equilibrium, in the limit of an infinitesimal output nudge, the converged prediction errors match the backprop errors up to a global sign set by the error convention and nudging direction. With the convention above, this can be written as $\varepsilon^{(l)} \to -\boldsymbol{\delta}^{(l)}$, so that the *local* energy-gradient update $\Delta\mathbf{W}^{(l)} \propto \varepsilon^{(l)} f\big(r^{(l-1)}\big)^\top$ recovers the supervised gradient-descent update $\Delta\mathbf{W}^{(l)} \propto -\boldsymbol{\delta}^{(l)}\mathbf{a}^{(l-1)\top}$ from part (a). State in one sentence why this is striking. (A full proof is long; give the key step — match the inference fixed-point equations to the backprop recursion of (a).)

(d) *Weight transport is inherited, not solved.* Observe that the relaxation in (b) still routes errors downward through $\mathbf{W}^{(l+1)\top}$. Explain, at the level of individual synapses, why requiring the feedback pathway to use the exact transpose of the feedforward weights is implausible.

(e) *Feedback alignment (Lillicrap et al., 2016).* Replace each transposed feedback matrix $\mathbf{W}^{(l+1)\top}$ by a *fixed random* matrix $\mathbf{B}^{(l+1)}$, giving the surrogate signal $\tilde{\boldsymbol{\delta}}^{(l)} = f'(\mathbf{z}^{(l)}) \odot \big(\mathbf{B}^{(l+1)} \tilde{\boldsymbol{\delta}}^{(l+1)}\big)$. Use it as a surrogate gradient direction, $\mathbf{U}^{(l)} = \tilde{\boldsymbol{\delta}}^{(l)} \mathbf{a}^{(l-1)\top}$, with weight update $\mathbf{W}^{(l)} \leftarrow \mathbf{W}^{(l)} - \eta \mathbf{U}^{(l)}$. Why would one naively expect random feedback to fail?

(f) *Descent condition.* For a single example (or in expectation over a minibatch), use $\Delta\mathcal{L} \approx -\eta \sum_l \big\langle \partial\mathcal{L}/\partial\mathbf{W}^{(l)},\, \mathbf{U}^{(l)} \big\rangle$ to show that the surrogate update of (e) still decreases the loss as long as the surrogate signal has positive overlap with the true error at each updated layer, i.e. $\boldsymbol{\delta}^{(l)\top} \tilde{\boldsymbol{\delta}}^{(l)} > 0$ (the angle to the true gradient is acute).

(g) *Alignment.* Describe the empirical mechanism that makes the condition in (f) hold during training: the forward weights $\mathbf{W}^{(l+1)}$ evolve so that the signal sent by $\mathbf{W}^{(l+1)\top}$ becomes approximately aligned with the signal sent by the fixed matrix $\mathbf{B}^{(l+1)}$. In this sense, $\mathbf{B}$ functions as a fixed, approximate "soft transpose." Why does this circumvent the weight-transport problem of (d)?

(h) *Synthesis.* Combining (c)–(g), sketch how a predictive-coding network with *separate* forward weights $\mathbf{W}^{(l)}$ and feedback weights $\mathbf{B}^{(l)}$ can approximate backprop without ever imposing $\mathbf{B} = \mathbf{W}^\top$ a priori. You may discuss either fixed random feedback (feedback alignment) or feedback weights learned by their own local rule. What gap remains between this scheme and exact gradient descent?

Part 3 — Reinforcement Learning & the Brain

Topics: Model-free RL: value, the Bellman equation, TD learning, and dopamine as a reward-prediction error; the actor–critic and the three-factor rule. Model-based RL: world models and planning, the successor representation, the hippocampal predictive map, replay/Dyna, and arbitration. Several parts are deliberately open-ended.

Problem 1. Model-free RL: value, prediction error, and the dopamine hypothesis [27 pts]

The value of a state is $V(s) = \mathbb{E}\big[\sum_{k\geq 0} \gamma^k r_{t+k} \mid s_t = s\big]$ with $\gamma \in [0,1)$, and the TD error is $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$ with update $V(s_t) \leftarrow V(s_t) + \alpha\, \delta_t$.

(a) *Bellman and discounting.* Derive $V(s) = \mathbb{E}\big[r + \gamma V(s')\big]$ from the definition, stating where the Markov property enters. On a deterministic linear track $s_1 \to s_2 \to \cdots \to s_N \to$ (terminal) with reward R only on entering the terminal state, solve for $V(s_i)$ in closed form and show value decays geometrically with distance to reward. Comment briefly on what γ means behaviorally (temporal discounting / impulsivity).

(b) *TD as an error signal.* Show that $\mathbb{E}[\delta_t \mid s_t = s] = 0$ at the true value function, so V is the fixed point of the update. For the linear track (initialize $V \equiv 0$, $\alpha = 1$), run TD by hand for the first two episodes and show that value propagates backward one state per episode from the reward — a concrete instance of *temporal credit assignment*.

(c) *Dopamine as a reward-prediction error.* Using a complete-serial-compound (tapped-delay-line) representation of the cue–reward interval and the same δ_t, account for Schultz et al.'s three canonical recordings: (i) an unpredicted reward, (ii) a fully predicted reward after learning, and (iii) omission of a predicted reward. Why is the omission *dip* below baseline time-locked to the expected reward time rather than to the (absent) reward?

(d) *Actor–critic and the three-factor rule.* In the basal ganglia a critic (ventral striatum) learns V and an actor (dorsal striatum) learns a policy, both taught by the dopaminergic δ. Explain how $\delta > 0$ vs. $\delta < 0$ act in opposite directions on the D1 ("Go") and D2 ("NoGo") pathways. Then, for the three-factor rule $\dot{w}_{ij} = \eta\, z_{ij}(t)\, \delta(t)$ with Hebbian eligibility trace $\dot{z}_{ij} = -z_{ij}/\tau_e + f(\text{pre}, \text{post})$, write the total change over a trial as $\Delta w_{ij} = \eta \int z_{ij}(t)\delta(t)\, dt$. Explain how the eligibility trace, which is itself a filtered memory of recent pre–post coincidences, solves the *distal-reward* problem when reward arrives seconds after the responsible coincidence (Yagishita et al., 2014, ∼1–2 s tag).

(e) *Open-ended — the backward shift of the dopamine signal.* TD with a complete-serial-compound state predicts that, across learning, the positive prediction error should migrate *gradually* backward in time, passing through intermediate time points between reward and cue. Address, in turn: **(i)** state precisely, from δ_t, why TD predicts this intermediate backward propagation, relating it to your episode-by-episode calculation in (b); **(ii)** this gradual shift was for years hard to observe — many studies reported that the response seemed to *appear* at the cue rather than travel there, while Amo et al. (2022) reported evidence for a gradual shift in a long-delay task; explain why the prediction is hard to test and what the choice of state representation (complete serial compound vs. microstimuli vs. a belief state under timing uncertainty) has to do with whether one expects a gradual vs. a "jumping" shift; **(iii)** propose at least one alternative account (e.g. distributional TD, belief-state TD, or value as a learned function of elapsed time) and say how it would change the predicted dopamine dynamics; **(iv)** design an experiment — recordings plus a task manipulation — that would discriminate a complete-serial-compound TD account from your alternative, and state which single measurement would be decisive.

Problem 2. Model-based RL: planning, the successor representation, and the cognitive map [24 pts]

Model-based RL learns a world model — transition probabilities $P(s' \mid s, a)$ and, here, state rewards $r(s)$ — and computes value by simulating futures, in contrast to the value caching of Problem 1.

(a) *Two systems.* For state rewards, write the model-based one-step look-ahead expression $Q(s,a) = \sum_{s'} P(s' \mid s,a)\big[r(s') + \gamma \max_{a'} Q(s',a')\big]$, and contrast model-based with model-free control along three axes: flexibility when rewards or transitions change, computational cost at decision time, and the implicated neural systems (e.g. prefrontal / hippocampal vs. dorsolateral striatal). Which corresponds to "goal-directed" and which to "habitual" behavior?

(b) *Successor representation.* Define the successor representation $M(s,s') = \mathbb{E}\big[\sum_{k\geq 0} \gamma^k \mathbf{1}(s_{t+k} = s') \mid s_t = s\big]$ and show that value factorizes as $V(s) = \sum_{s'} M(s,s')\, r(s')$, i.e. $\mathbf{V} = \mathbf{M}\,\mathbf{r}$. Give a TD-style learning rule for M (treat the indicator $\mathbf{1}(s_t = \cdot)$ as the "reward"). Explain why the SR is an intermediate between model-free and model-based control: to which kind of change — in the reward function or in the transition structure — can it adapt *instantly*, and to which can it not?

(c) *The hippocampal predictive map.* It has been proposed that hippocampal place fields encode the SR (a predictive map) and that grid cells relate to its low-dimensional eigenstructure (Stachenfeld et al., 2017). State one concrete, testable prediction this makes about how place fields should *deform* in a structured environment — e.g. near a barrier, or under a directed vs. random foraging policy. *Open-ended:* design an experiment that would distinguish an SR-based predictive map from a purely Euclidean cognitive map.

(d) *Open-ended — replay, Dyna, and arbitration.* Dyna interleaves real experience with model-based updates computed from *simulated* experience, and hippocampal replay is a candidate neural substrate (tying back to the consolidation lectures). Address: **(i)** explain how replaying remembered or simulated transitions lets a model-free learner approach model-based performance without running an explicit forward model at decision time; **(ii)** given that model-based control is flexible but costly while model-free control is cheap but inflexible, propose a principle by which the brain should *arbitrate* between them, and predict when each should dominate (consider amount of training, reward stability, time pressure, and reward devaluation); **(iii)** sketch an experiment — e.g. a two-step task with reward devaluation — whose behavioral signature would reveal which system is in control.

Instructors: numerical parts are light by design and can be expanded into coding assignments — e.g. Oja's rule on natural-image patches; sparse coding to recover Gabor-like dictionaries; TD on a gridworld with a simulated dopamine read-out; learning a successor representation and visualizing the resulting predictive place fields; or a Dyna agent on a changing-reward task. The open-ended parts of Problem Set 3 are intended for short written discussion and can double as exam or seminar prompts. A separate solution key can accompany this set.